

Endomorfizmi vekt. pr.

① $A: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ v std. bazi pripade matrika $A = \begin{bmatrix} 3 & 2 & 2 \\ 1 & 4 & 1 \\ -2 & -4 & -1 \end{bmatrix}$

Izračunaj lastne vrednosti & lastne podprostore A in ga diagonaliziraj.

$$\begin{aligned} p(\lambda) = \det(A - \lambda I) &= \det \begin{bmatrix} 3-\lambda & 2 & 2 \\ 1 & 4-\lambda & 1 \\ -2 & -4 & -1-\lambda \end{bmatrix} = (3-\lambda)(4-\lambda)(-1-\lambda) - 4 - 8 - 2(1-\lambda) \\ &\quad - (3-\lambda)(-4) - 2(-2)(4-\lambda) = \\ &= -(12 - 7\lambda + \lambda^2)(1+\lambda) - 12 + 2 + 2\lambda + 12 - 4\lambda + 16 - 4\lambda = \\ &= -\lambda^3 + 7\lambda^2 - \lambda^2 + 7\lambda - 12\lambda - 12 + 2 - 6\lambda + 16 = \\ &= -\lambda^3 + 6\lambda^2 - 11\lambda + 6 = (\lambda-1)(-\lambda^2 + 5\lambda - 6) = -(\lambda-1)(\lambda-3)(\lambda-2) = \underline{-(1-\lambda)(3-\lambda)(2-\lambda)} \end{aligned}$$

	-1	6	-11	6
1		-1	5	-6
	-1	5	-6	0

$\lambda_1 = 1$
 $\lambda_2 = 3$
 $\lambda_3 = 2$

 Lastne vrednosti

$$\lambda_1: A - \lambda_1 I = \begin{bmatrix} 2 & 2 & 2 \\ 1 & 3 & 1 \\ -2 & -4 & -2 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 0 \\ 0 & 2 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \begin{array}{l} x_2 = 0 \\ x_1 = -x_3 \end{array}$$

$$E_1(1) = \text{Span}\{(1, 0, -1)\}$$

$\parallel$
 v_1

$$\lambda_2: A - \lambda_2 I = \begin{bmatrix} 0 & 2 & 2 \\ 1 & 1 & 1 \\ -2 & -4 & -4 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 2 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \quad \begin{array}{l} x_2 + x_3 = 0 \\ x_1 = 0 \end{array}$$

$$E_2(3) = \text{Span}\{(0, 1, -1)\}$$

$\parallel$
 v_2

$$\lambda_3: A - \lambda_3 I = \begin{bmatrix} 1 & 2 & 2 \\ 1 & 2 & 1 \\ -2 & -4 & -3 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 2 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} \quad \begin{array}{l} x_3 = 0 \\ x_1 + 2x_2 = 0 \\ x_1 = -2x_2 \end{array} \quad E_3(2) = \text{Span}\{(-2, 1, 0)\}$$

$\parallel$
 v_3

$$A(v_1) = \lambda_1 v_1 = v_1$$

$$A(v_2) = \lambda_2 v_2 = 3v_2$$

$$A(v_3) = \lambda_3 v_3 = 2v_3$$

$$\mathcal{B} = \{v_1, v_2, v_3\}$$

$$[A]_{\mathcal{B}}^{\mathcal{B}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$[A]_{\mathcal{E}}^{\mathcal{E}} = [id]_{\mathcal{E}}^{\mathcal{B}} [A]_{\mathcal{B}}^{\mathcal{B}} [id]_{\mathcal{B}}^{\mathcal{E}} = P D P^{-1}$$

$$P = \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 1 \\ -1 & -1 & 0 \end{bmatrix}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ -1 & -1 & 0 & 0 & 0 & 1 \end{array} \right] \sim \left[\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & -1 & -2 & 1 & 0 & 1 \end{array} \right] \sim$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 \end{array} \right] \sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 3 & 2 & 2 \\ 0 & 1 & 0 & -1 & 0 & -1 \\ 0 & 0 & 1 & 1 & 1 & 1 \end{array} \right]$$

$$P^{-1} = \begin{bmatrix} 3 & 2 & 2 \\ -1 & 0 & -1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$[A]_{\mathcal{E}}^{\mathcal{E}} = \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 1 \\ -1 & -1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} 3 & 2 & 2 \\ -1 & 0 & -1 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 2 & 2 \\ 1 & 4 & 1 \\ -2 & -4 & -1 \end{bmatrix}$$

$$(2.) T: \mathbb{R}_2[x]^{\mathcal{D}}$$

$$T(x \overset{p_1}{+} 1) = -2x^2 + x + 3$$

$$T(x^2 \overset{p_2}{+} 1) = -3x - 3$$

$$T(x^2 \overset{p_3}{+} x + 1) = x^2 + x + 1$$

l.zr. lastne vrednosti, lastne podpr. T

$$\mathcal{B} = \{p_1, p_2, p_3\}$$

$$1 = p_1 + p_2 - p_3$$

$$x = p_3 - p_2$$

$$x^2 = p_3 - p_1$$

$$[id]_{\mathcal{B}}^{\mathcal{E}} = \begin{bmatrix} 1 & 1 & -1 \\ 0 & -1 & 1 \\ -1 & 0 & 1 \end{bmatrix}$$

$$T(p_1) = -2(p_3 - p_1) + (p_3 - p_2) + 3(p_1 + p_2 - p_3) = 5p_1 + 2p_2 - 4p_3$$

$$T(p_2) = -3(p_3 - p_2) - 3(p_1 + p_2 - p_3) = -3p_1$$

$$T(p_3) = (p_3 - p_1) + (p_3 - p_2) + (p_1 + p_2 - p_3) = p_3$$

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{bmatrix} 5 & -3 & 0 \\ 2 & 0 & 0 \\ -4 & 0 & 1 \end{bmatrix}$$

$$\det(T - \lambda I) = \det \begin{bmatrix} 5-\lambda & -3 & 0 \\ 2 & -\lambda & 0 \\ -4 & 0 & 1-\lambda \end{bmatrix} = (5-\lambda)(-\lambda)(1-\lambda) - (-3)(2)(1-\lambda) =$$

$$= (1-\lambda)(\lambda^2 - 5\lambda + 6) = (1-\lambda)(\lambda-2)(\lambda-3) \rightarrow \boxed{\begin{matrix} \lambda_1 = 1 \\ \lambda_2 = 2 \\ \lambda_3 = 3 \end{matrix}} \text{ NEODVISNE OD BAZE}$$

$$\lambda_1: E_T = \text{Span}\{p_3\} \\ v_1 = p_3$$

$$\lambda_2:$$

$$T - 2I = \begin{bmatrix} 3 & -3 & 0 \\ 2 & -2 & 0 \\ -4 & 0 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & -4 & -1 \end{bmatrix}$$

$$\begin{matrix} x_1 = x_2 \\ 4x_2 = -x_3 \\ x_3 = -4x_2 \end{matrix} \rightarrow E_T(2) = \text{Span}\{(1, 1, -4)\} \\ v_2 = (1, 1, -4) = p_1 + p_2 - 4p_3$$

$$\lambda_3:$$

$$T - 3I = \begin{bmatrix} 2 & -3 & 0 \\ 2 & -3 & 0 \\ -4 & 0 & -2 \end{bmatrix} \sim \begin{bmatrix} 2 & -3 & 0 \\ 0 & 0 & 0 \\ 0 & -6 & -2 \end{bmatrix}$$

$$2x_1 = 3x_2$$

$$3x_2 = -x_3$$

$$\left. \begin{matrix} x_3 = -3x_2 \\ x_2 = x_2 \\ x_1 = \frac{3}{2}x_2 \end{matrix} \right\} E_T(3) = \text{Span}\{(3, 2, -6)\} \\ v_3 = (3, 2, -6) = 3p_1 + 2p_2 - 6p_3$$

$$T_{\mathcal{B}'}^{\mathcal{B}'} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

$$\mathcal{B}' = \{v_1, v_2, v_3\}$$

3.) Ali sta endomorfizma diagonalizabilna?

a) $A: \mathbb{R}^2 \rightarrow \mathbb{R}^2 \quad A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

b) $A: \mathbb{R}^3 \rightarrow \mathbb{R}^3 \quad A = \begin{bmatrix} 3 & 2 & 1 \\ 0 & 3 & 1 \\ 0 & 0 & 3 \end{bmatrix}$

A je diagonalizabilen, če obstaja baza iz lastnih vektorjev ($\text{akr}(\lambda_i) = \text{gkr}(\lambda_i)$ za vsak i)

$$E_{\lambda_i}(\lambda_i) = \text{gkr}(\lambda_i)$$

$\text{akr}(\lambda_i) \geq \text{gkr}(\lambda_i) \geq 1$
stopnja ničeln polinom

a) $p_A = (\lambda + 1)(\lambda - 1)$

$\text{gkr}(\lambda_i) = \text{akr}(\lambda_i) = 1$ za $\forall i \Rightarrow A$ je diag.

b) $p_A = (3 - \lambda)^3$

$$(A - 3I) = \begin{bmatrix} 0 & 2 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \text{rk}(A - 3I) = 2$$

$$\dim(\ker(A - 3I)) = 1 = \text{gkr}(3) \Rightarrow \text{gkr}(3) \neq \text{akr}(3) \Rightarrow A \text{ ni diag.}$$

4.) za katere $t \in \mathbb{R}$ se da diag.

$$A = \begin{bmatrix} 1 & t & 25 \\ 0 & t & t+1 \\ 0 & 0 & -1 \end{bmatrix}$$

$$p_A = (1 - \lambda)(t - \lambda)(-1 - \lambda)$$

$$\lambda_1 = 1$$

$$\lambda_2 = t$$

$$\lambda_3 = -1$$

① $t \neq 1 \wedge t \neq -1$

$\text{gkr}(\lambda_i) = \text{akr}(\lambda_i) = 1$ za $\forall i \Rightarrow A$ diag.

② $t = 1$:

$$A - I = \begin{bmatrix} 0 & 1 & 25 \\ 0 & 0 & 2 \\ 0 & 0 & -2 \end{bmatrix}$$

$$\text{akr}(-1) = 1 = \text{gkr}(-1) \checkmark$$

$$\text{akr}(1) = 2$$

$$\text{rk}(A - I) = 2 \rightarrow \dim(\ker(A - I)) = 1 = \text{gkr}(1)$$

$\Rightarrow A$ ni diag.

③ $t = -1$:

$$\text{akr}(1) = 1 = \text{gkr}(1) \checkmark$$

$$\text{akr}(-1) = 2$$

$$A + I = \begin{bmatrix} 2 & -1 & 25 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\text{rk}(A + I) = 1 \rightarrow \dim(\ker(A + I)) = 2 = \text{gkr}(-1) = \text{akr}(-1)$$

$\Rightarrow A$ je diag.

$\Rightarrow A$ je diag. $\Leftrightarrow t \in \mathbb{R} \setminus \{1\}$

